MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 7768

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, metal-ceramic

See figure 1

Mounting position: Any

Weight: 0.2 ounce (5.67 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ik	Pp	TE	Barometric pressure, reduced
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	W	°C	mmHg
Maximum:	6.6	330	---	50	---	0.01	30	5.5	250	8
Minimum:	6.0	---	-50	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	200	+6	---	270	---	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(D)	Qualification inspection								
1002	Barometric pressure, reduced	-	Pressure = 8 +1 mmHg; voltage = 300 Vac between grid and anode	---	---	---	---	---	---
1031	Sweep-frequency vibration	6	Ebb = 250 Vdc; Ec = 0; Rk = 68 ohms; Rp = 2,000 ohms	---	---	Ep	---	75	mVac
1031	Sweep-frequency vibration fatigue	8	Eb = Ec = 0; Ef = 6.3 V	---	---	---	---	---	---
---	Sweep-frequency vibration fatigue test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	75	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	30	µAdc
1306	Transconductance (1)	-		---	---	Sm	32,500	---	µmhos
1301	Heater current	-		---	---	If	370	440	mA
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	370	430	mA
1336	Heater-cathode leakage	-		0.65	II	Ihk	---	20	µAdc
1266	Total grid current	1	Eb = 300 Vdc; Ecc = -0.8 Vdc; Pc = 0.1 Mec; Rk = 22 ohms	0.65	II	Ic	---	-1.0	µAdc
1256	Electrode current (anode)	-	Ec = 0; Rk = 22 ohms	0.65	II	Ib	14	30	mAdc
1306	Transconductance (1)	2		0.65	II	Sm	40,000	60,000	µmhos
1261	Electrode voltage (grid)	-	Rk = 0; Ec/Ib = 0.1 mAdc	0.65	II	Ec	---	-5.0	Vdc
---	Noise figure	3	Ebb = 280 Vdc; Ec = 0; Rb = 3,300 ohms; Cb = 10 µF; Rk = 22 ohms	0.65	II	NF	---	4.5	dB
1201	Short and discontinuity detection	-		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	-		---	---	R	50	---	Meg
1306	Transconductance (2)	2	Ef = 5.7 V	---	---	ΔSm Ef	---	20	%
1316	Amplification factor	2		---	---	Mu	180	270	---
1266	Grid emission	4	Ef = 7.0 V; Ecc = -15 Vdc; Rk = 0; Rg = 0.1 Meg	---	---	Ic	0	-2.0	µAdc
2210	Grid recovery	7	Ebb = 250 Vdc; Ec/Ib = 10 mAdc; Rb = 0.01 Meg	---	---	ΔIb Δib	--- ---	1.0 2.5	mAdc ma

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
3131	Direct-interelectrode capacitance	-		—	—	Cgp Cin Cout Chk	1.3 4.5 0.010 1.5	2.1 7.5 0.026 3.3	pF pF pF pF
2126	Envelope strain	-		—	—	—	—	—	---
(D)1031	Low-frequency vibration	5	Ebb = 250 Vdc; Ec = 0; Rk = 68 ohms; Rp = 2,000 ohms; 15 G; F = 40 Hz	—	---	Ep	—	50	mVac
	Quality conformance inspection, part 3								
3506	Heater-cycling life	9	Ef = 7.0 V; Ehk = +70 Vdc; Rk = 0; Ec = Eb = 0	—	—	—	—	---	—
---	Heater-cycling life-test end point:								
1336	Heater-cathode leakage	-		—	—	Ihk	---	40	µAdc
1516	Stability life	-	Eb = 200 Vdc; Ecc = +7 Vdc; Ehk = -70 Vdc; Rk = 270 ohms; Rg = 0.01 Meg	2.5	Code F	—	---	—	—
---	Stability life-test end point:								
---	Change in transconductance (1) of individual tubes	-		—	—	ΔSm t	—	15	%
1501	Intermittent life	-	Group A; stability life-test conditions, or equivalent; t = 1,000 hours	—	---	—	---	---	—
---	Intermittent life-test end points:								
1266	Total grid current	-		—	---	Ic	---	-1.0	µAdc
1301	~~Heater current~~	-		—	---	If	370	430	mA
1306	Transconductance (1)	-		—	---	Sm	32,500	—	µmhos
—	~~Change in transconductance~~ (1) of individual tubes			—	---	ΔSm t	—	30	%
1306	Transconductance (2)	-		—	—	ΔSm Ef	—	30	%
---	Noise figure	-		—	—	NF	—	6.0	dB
1336	Heater-cathode leakage	-		—	—	Ihk	—	20	µAdc
1211	Insulation of electrodes	-		—	—	R	20	—	Meg
(D)1041	Shock	10	~~Hammer~~ angle = 30°; Eb = 250 Vdc; Ec = 0; Rk = 68 ohms; Ehk = +100 Vdc	—	—	—	—	—	---
---	Shock test end points:								
1031	Low-frequency vibration	10		—	—	Ep	—	75	mVac

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
	Shock test end points: -Continued								
1336	Heater-cathode leakage	10		---	---	Ihk	---	30	uAdc
1306	Transconductance (1)	10		---	---	Sm	32,500	---	umhos
1301	Heater current	10		---	---	If	370	440	mA

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. In performing this test, the cathode bypass capacitor should be charged to +6 volts prior to insertion of the tube into the test socket to prevent high initial surge current.
3. Noise figure test:
   a. Definition. Noise figure shall be defined as:

      $$F = 10 \log \frac{Si/Ni}{So/No}$$ where

      F is noise figure expressed in dB.
      Si/Ni is signal to noise ratio at input of tube under test.
      So/No is signal to noise ratio at output of tube under test.

   b. Test procedure. Tube under test shall be operated in a grounded grid amplifier circuit at 200 ±10 MHz. The input circuit shall be adjusted for a noise match and tuned for a minimum noise figure. The output circuit shall be tuned and adjusted for a power match. External feedback around the tube under test except for that required for neutralization or that which is unavoidable in a well designed circuit, shall not be used.

      A signal source which has known output constants over the bandwidth of the tube under test, such as a noise diode, gas discharge tube, or known resistance at a known temperature shall be used in measuring the noise figure of the tube under test.

      The bandwidth of the tube under test circuit shall be at least twice that of the overall system used in measuring noise figure.
4. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. Test within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ecc Vdc	Ebb Vdc	Rk ohms	Rb Meg	Rg Meg
7.0	0	325	50	0.003	0.18

5. The tube shall be vibrated with simple harmonic motion in each of two planes; (1) parallel to the cylindrical axis; (2) perpendicular to the cylindrical axis and parallel to a line through the heater pins.
6. The sweep-frequency vibration test shall be performed as follows:
   a. The frequency shall be increased from 100 to 2,000 Hz with approximately logarithmic progression in 3 ±1 minutes. The return sweep (2,000 to 100 Hz) is not required.
   b. The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and perpendicular to a line through the heater pins.
   c. The peak acceleration shall be maintained at 10 ±1.0 G throughout the test.
   d. The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of the vibration shall be measured with a suitable device having a response to the rms value of the voltage to within ±0.5 dB of the response at 400 Hz for the frequency range of 100 to 3,000 Hz, and having a bandpass filter with an attenuation rate of 24 dB per octave below the low frequency cut-off point of 50 Hz and above the high frequency cut-off point of 5,000 Hz. The meter shall have a dynamic response characteristic equivalent to or faster than a VU meter (operated in accordance with Standard ASA No. C 16.5-1954).

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NOTES: -Continued

7. Test shall be performed as follows:

   a. With TUT operating under specified conditions, adjust Ec for Ib = 10 mAdc.

   b. Apply 3-volt pulse, 20 µs duration, 60 pps to the grid.

   c. With application of pulse, measure undershoot (Δib) and change in average current (ΔIb).

8. Sweep-frequency vibration fatigue. Test shall be performed as specified in method 1031, except that the tubes shall be vibrated for a total of 6 hours, that is, 3 hours in each of two directions; (1) parallel to the cylindrical axis; (2) perpendicular to the cylindrical axis and perpendicular to a line through the heater pins.

9. Heater-cycling life-test sampling procedure and acceptance criteria shall be as follows: First sample (n1) shall consist of 15 tubes with an acceptance number (c1) of zero (0). Second sample (n2) shall consist of 15 tubes with an acceptance number (cc) equal to one (1) for the cumulative sample.

10. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - ME
Air Force - 17, 99
DLA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Army - EL

Agent:
DLA - ES

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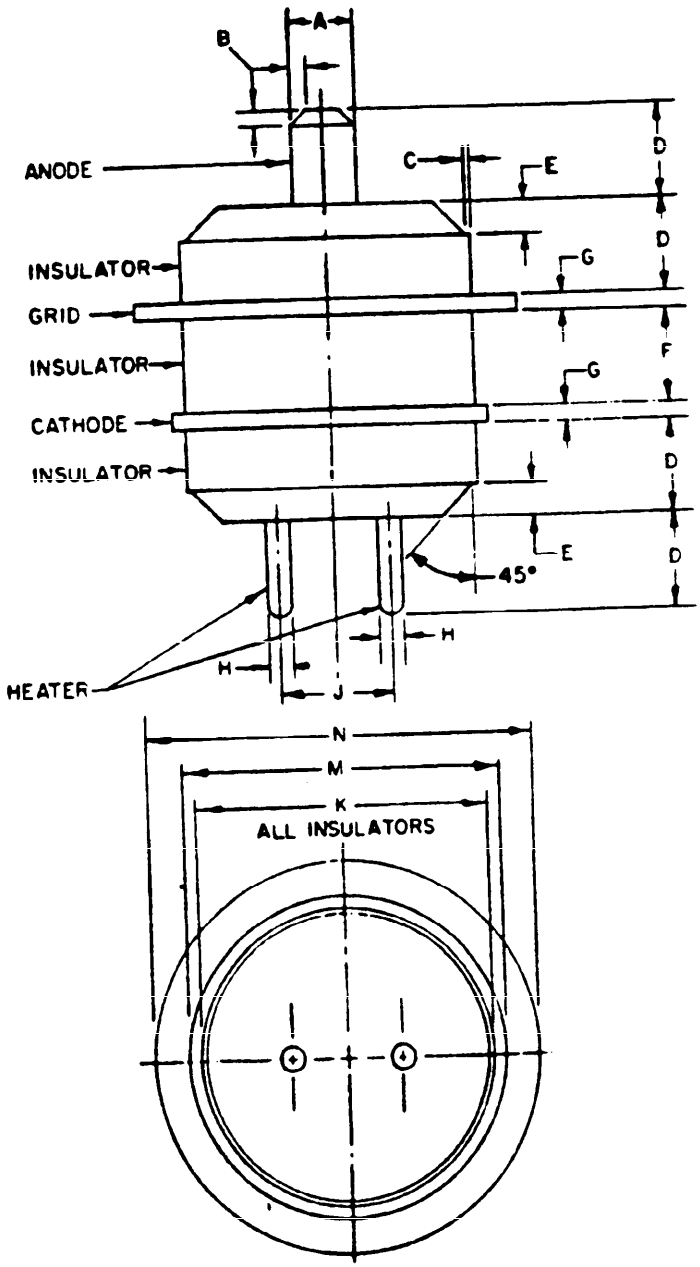


FIGURE 1. Outline drawing of electron tube type 7768.

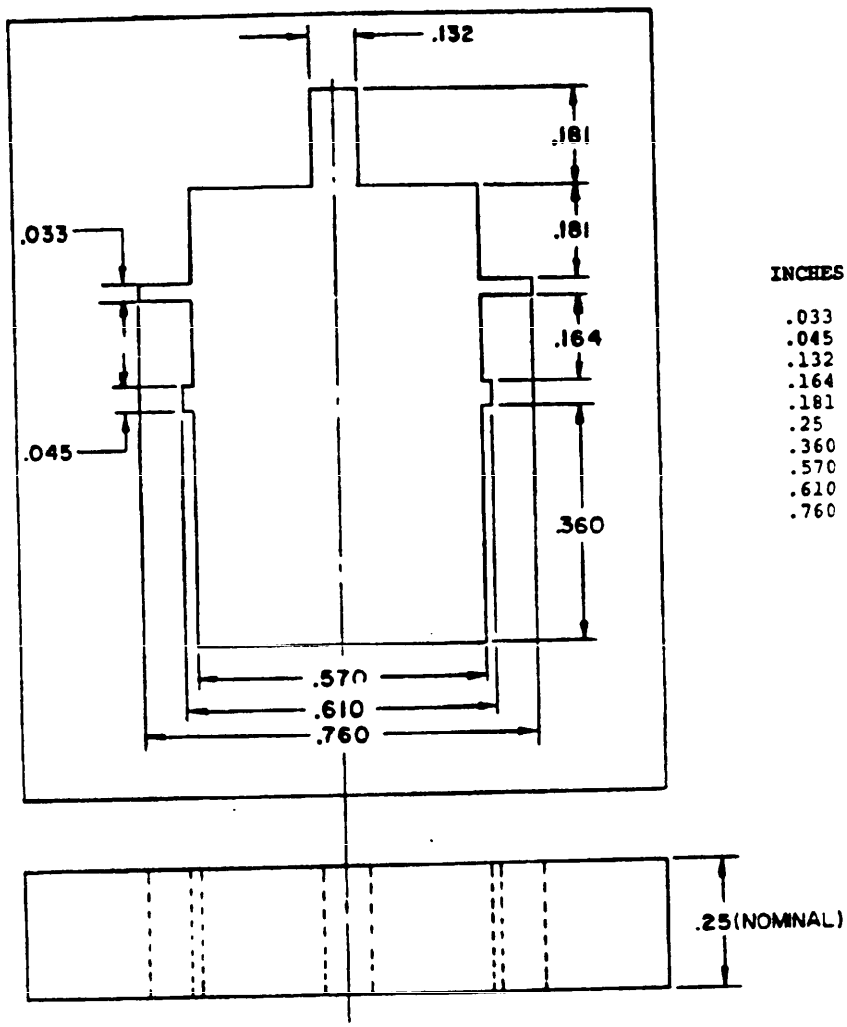
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note)		
A	.122 (3.10)	.128 (3.25)
F	.167 (4.24)	.177 (4.50)
J	.185 (4.70)	.215 (5.46)
M	.598(15.19)	.608(15.44)
N	.748(19.00)	.758(19.25)
Quality conformance inspection, part 2		
D	.170 (4.32)	.180 (4.57)
E	.040 (1.02)	.060 (1.52)
G	.025 (.64)	.031 (.79)
H	.047 (1.19)	.053 (1.35)
K	.535(13.59)	.565(14.35)
Reference dimensions		
B	.030(.76)	
C	.005(.13)	

NOTE: The AQL for dimensions listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

FIGURE 1. Outline drawing of electron tube type 7768 - Continued.

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INCHES	MM
.033	.84
.045	1.14
.132	3.35
.164	4.17
.181	4.60
.25	6.35
.360	9.14
.570	14.48
.610	15.49
.760	19.30

NOTES:
a. Dimensions are in inches.
b. Tolerances of all gage dimensions are ±.001(.03 mm) inches.

FIGURE 2. Alinement gage.

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	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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